O-C-230

MAY 6, 1958

(See 6.3)

FEDERAL SPECIFICATION

CHARGE, FIRE EXTINGUISHER, FOAM

This specification was approved by the Commissioner, Federal Supply Service, Gcneral Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers charges for chemical foam-type fire extinguishers and generators. They are for use in portable and wheeled extinguishers for extinguishing oil and gasoline fires. For detailed intended use see 6.1.

1.1.1 *Federal specification* coverage.—Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specification, but include only those most generally used by the Federal Government.

1.2 C1assification.

1.2.1 *Types and* sizes.—The charges shall be of the following types and sizes, as specified (see 6.2):

Type I.—Two-component charge for chemical foam extinguishers.
- Size l.—For nominal 2-1/2-gallon foam extinguisher.
- Size 2.—For nominal 40-gallon foam extinguisher.

Type III. — Single-component charge for chemical foam generators (single hopper.

Type VII. — Two-component charge for chemical foam generators (dual hopper).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standands-The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L-P406 — Plastics, Organic, General Specifications, Test Methods.

O-F-361 — Fire-Extinguishers; Hand, Portible, Foam-Type.

O-S-576 - Sodium-Bicarbonate, Technical Grade.

QQ-S-766 — Steel, Corrosion-Resisting, Plates, Sheets, Strips, and Structural Shapes.

UU-P-31-Paper; General Specifications and Methods of Testing.

ZZ-H-451-Hose; Fire; Cotton, Rubber-lined.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591-Boxes, Fiberboard, Wood-Cleated.

PPP-B-601-Boxes, Wood, Cleated-Plywood .

PPP-B-621-Boxes, Wood, Nailed and Lock-Comer.

PPP-B-636-&Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102--Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which in. cludes cumulative monthly suppliements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Admin-

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istration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-F-482 — Fire Engines, Chemical, Foam Type.

MIL-G-636-Generators, Foam, Hopper-feed, Fire Fighting, Chemical Type.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129— Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking for U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Nongovernmental.

Consolidated Freight Classificaiton Rules and Container Specifications.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

American Trucking Association Publication:

National Motor Freight Classification Rules and Container Specification.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N. W., Washington, D. C.)

3. REQUIREMENTS

3.1 Description. — The chemical foam charges shall consist of aluminum sulfate and sodium bicarbonate as the chemically active constituents. Foaming agents, foam stabilizers and, if necessary, preservatives and fluidizing material shall be used. The charges shall not cake or deteriorate in the container.

3.1.1 *Types 1 and* VII.—Types I and VII shall consist of two components. Component A shall be aluminum sulfate. Component B shall consist of a homogeneous mixture of sodium bicarbonate, suitable foaming agents and foam stabilizers, and if necessary pre-

TABLE I.—*Active components on an anhydrous basis, weight requirements, and water volumes required for mixing*

Type	Size	"A" Component			"B" Component		
		Aluminum sulfate	(min.)	Volume of water required	Sodium bicarbonate	(min.)	Volume of water required
		Lb.	*Oz.*	*Gal.*	*Lb.*	*Oz.*	*Gal.*
I	1	1	--	⅝ at 150° F (approx.)	1	6	1¾ at 100° F (approx.)
	2	18	8	5¼ at 150° F (approx.)	24	--	29 at 100° F (approx.)
VII [1]	--	28	8	----	39	--	----

[1] The minimum total net weight of the two components as packaged shall be 100 pounds.

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servatives and fluidizing materials. The chemically active constituents shall be supplied in the amounts shown in table I for the type and size required.

3.1.2 Type *III*.—Type III charge shall consist of a single component which shall be an intimate mixture of sodium bicarbonate (approximately 99 percent pure), sufficient quantity of aluminum sulfate to react with all of the sodium bicarbonate (substantially according to the chemical equation for such reaction), and a foam stabilizer in sufficient quantity to produce a strong, closely-knit, heat-resistant bubble structure. Chemically inert material in the dry powder condition to act *as* inhibitors may be added to produce rigidity and resistance to heat; the net combined weight of the blended components shall be no less than 50 pounds.

3.2 Preproduction sample.-When specified, a preproduction sample shall be furnished. The preproduction sample is defined as a production line sample of the product which the manufacturer will subsequently produce in fulfillment of the contract, As soon as practicable after award of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a preproduction sample, when specified. for examination and test to determine conformance to this specification. Approval of the preproduction sample by the activity concerned shall not relieve the contractor of his obligation to supply material conforming to this specification. Any changes or deviations from the preproduction sample shall be subject to the approval of the contracting officer.

3.3 Material.—Material shall be as specified herein. Material not definitely specified shall be of a good commercial quality for this type of commodity. Material shall be nontoxic under normal conditions of use. When mixed with the required amount of water, charges shall not corrode the extinguisher.

3.3.1 *Aluminum* sulfate.-Aluminum sulfate shall be of the technical grade and stable hydrate suitable for the purpose, free of acidic or basic impurities, and prepared so as to be free flowing and easily dissolved. Crystalline aluminum sulfate shall contain no more than 0.5 percent by weight water-insoluble matter and no more than 0.75 percent by weight iron as Fe_2O_3. These properties shall be maintained during storage periods specified. The aluminum sulfate shall be of such fineness as to pass through 16 mesh sieve; no more than 30 percent shall be retained on 60 mesh sieve, and no less than 50 percent shall be retained on 100 mesh sieve.

3.3.2 *Sodium* bicarbonate.-Sodium bicarbonate shall conform to Federal Specification O-S-576.

3.4 Foam characteristics. -When the chemicals are mixed with the required amount of water for the type of charge used, the foam produced on discharge from the extinguisher shall be a stable, cohesive foam of a satisfactory degree of fluidity for extinguishing oil and gasoline fires. The foam shall consist of a mass of small bubbles not easily broken down by intense heat, shall adhere to solid surfaces, shall float on gasoline and oil surfaces, and shall not be injurious to persons or equipment.

3.5 Performance.

3.5.1 *Foam volume*.-When the charges are tested as specified in 4.3.5, the foam volume per pound of dry combined components A and B shall be as shown in table II.

TABLE II.—Foam volume

Type	Size	Yield per pound of combined components A and B
		Gallons
I	1 and 2	7.5
III	------	8.0
VII	------	7.5

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3.5.1.1 *Foam volume retraction.* — For types I and VII, the volume of foam shall not retract to less than 65 percent of the original volume after 15 minutes exposure to the atmosphere at plus 70° F.

3.5.2 Heat resistance.—When the charges are tested as specified in 4.3.6 and after the test fire has burned four minutes, the foam blanket shall cover not less than one-half the test pan fuel surface and the remaining foam shall provide a surface cover impermeable to fuel vapors. Foams producing a foam blanket which will reseal the gasoline surface and prevent the reignition of the gasoline for a period of four minutes will be considered acceptable.

3.5.3 *Type I, sizes 1 anti 2.*

3.5.3.1 *Extinguisher* pressure.-When the charges *are* tested as specified in 4.3.4, the pressure in the fire extinguisher shall be not greater than 350 p.s.i.g. at plus 120° F. with the extinguisher nozzle shut off. The greatest pressure shall be reached not sooner than five seconds after the initiation of the reaction. With the extinguisher nozzle open, the maximum pressure developed in the fire extinguisher shall be not less than 35 p.s.i.g.

3.5.3.2 *Discharge range. — When the charges* are tested as specified in 4.3.4 with the solution at 70° F., the foam from the type I, size 1 charge snail discharge a minimum horizontal distance of 20 feet within 10 seconds, maintain a range in excess of 20 feet for not less than 45 seconds, and maintain a range beyond a distance of 10 feet for at least 60 seconds from the time the extinguisher is actuated. When tested under the same conditions, the type I, size 2 charge shall discharge a minimum horizontal distance of 35 feet within 10 seconds, maintain a range in excess of 35 feet for not less than 3 minutes, and maintain a range beyond a distance of 20 feet for at least 4 minutes from the time the extinguisher is actuated.

3.5.3.3 *Fire extinguishment. — When* the charges are tested as specified in 4.3.3, the foam shall extinguish the fire. The depth of the foam shall be not less than 1 1/4 inches upon extinguishment of the fire.

3.6 Identification marking and instructions for use.—In addition to any special marking specified herein. metal containers may be provided with any supplementary marking which might be required for commercial use, by suitable lithographing on the reverse side of the marking as herein specified. Other identification marking, applicable to Department of Defense purchases, shall be in accordance with Military Standard MIL-STD-130.

3.6.1 *Type I.*

3.6.1.1 Size 1, *nominal 2 1/2-gallon foam extinguisher charge*.—Containers or bags containing components A and B shall bear labels with the following instructions printed thereon together with any other pertinent data. Directions for component A shall be black characters on yellow backgound: directions for component B shall be yellow characters on black background.

A-COMPONENT (FOR INNER CHAMBER)

Instructions for Use

(1) Wash inner chamber thoroughly.
(2) Using separate container, dissolve contents of this package in 2 1/4 pints of water at approximately 150° F. If hot water is not available, cold water may be used.
(3) Stir until completely dissolved and pour solution into inner chamber to filler mark. (If of insufficient volume. make up with water).
(4) Replace stopple on inner chamber.

E—COMPONENT (FOR OUTER CHAMBER)

Instructions for Use

(1) Wash outer chamber thoroughly.
(2) Run water through hose to make sure hose is unobstructed.

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(3) Using separate container, dissolve contents of this package in 1 3/4 U. S. gallons of water at approximately 100°F. (never to exceed 110°F.). If warm water is not available cold water may be used.
(4) Pour solution into outer chamber to filler mark. (If of insufficient volume; make up with water.)
(5) Place inner chamber in extinguisher.
(6) Screw cap on tight.
(7) Record date on tag attached to extinguisher.

3.6.1.1.1 *Marking* of boxes.—Boxes containing both A and B components shall be marked as follows:

"Type I charges, for 2 1/2-gallon foam extinguishers. Full directions are on containers. (Manufacturer's name) (Date of packaging) (Any special requirements for storage) (Control and batch number) ."

3.6.1.2 Size 2, *nominal 40-gallon foam extinguisher charge.* — Containers containing components A and B shall bear labels with the following instructions printed thereon together with any other pertinent data. Directions for component A shall be black characters on yellow background; directions for component B shall be yellow characters on black background.

A—COMPONENT (FOR INNER CHAMBER)

Instructions for Use

(1) Wash inner chamber thoroughly.
(2) Using separate container, dissolve contents of this package in 5 1/4 U. S. gallons of water at approximately 150°F. If hot water is not available cold water may be used.
(3) Stir until completely dissolved.
(4) After filling outer chamber and inserting inner chamber into extinguisher, pour solution into inner chamber to filler mark. (If of insufficient volume, make up with water.)
(5) Screw cap on tight.
(6) Close valve on cap.
(7) Record date on tag attached to extinguisher.

B-COMPONENT (FOR OUTER CHAMBER)

Instructions for Use

(1) Wash outer chamber thoroughly.
(2) Run water through hose to make sure hose is unobstructed.
(3) Using separate container, dissolve contents of this package in 29 U. S. gallons of water at approximately 100°F. (never to exceed 110°F.). If warm water is not available, cold water may be used.
(4) Pour solution into outer chamber to filler mark. (If of insufficient volume, make up with water.)
(5) Place inner chamber in extinguisher.

NOTE: Follow by steps (5), (6), and (7) for "A" component.

3.6.1.2.1 *Marking of boxes.—Boxes* containing both .4 and B components shall be marked as follows:

"Type I charges, for 20-gallon foam extingishers (single cylinders), Full directions are on containers. (Manufacturer's name) (Date of packaging) (Any special requirements for storage) (Control and batch number) ."

3.6.2 *Type III* .—Unless otherwise specified, the marking for identification and instructions for use on the type III chemical foam charge containers shall be in black letters and numerals against a red background and shall contain the information and be in the form shown on figure 1. The container shall be marked so that water immersion, contact with contents of the container, or normal handling shall not impair the legibility of the marking or remove the marking from the container. No paper labels shall be used.

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12 1/2"

THIS END UP

CHEMICAL

FOAM POWDER

TYPE III CHARGE

(Single Hopper Foam Generator)
In Accordance With

FEDERAL SPECIFICATION O-C-230

For Use In

Making Chemical Foam
For Fire Extinguishment

STORE IN A COOL DRY PLACE 11"

instructions for Use

This foam powder is for use with either installed or portable continuous foam generators (single-hopper type)

TO OPEN: Pull handle attached to the strip-type cover. Start water flowing through the ejector before emptying contents of this container into the hopper of the foam generator,

CAUTION

Make sure the powder is free flowing (not caked or lumpy) before emptying into the hopper. The powder should not be packed down but pushed easily from the sides into the center of the hopper.

MANUFACTURER'S NAME
ADDRESS LOCATION

Date of Packaging Batch No.

FIGURE l.—Marking for type III charge container.

3.6.3 Type VIl. .—The marking on the component .4 containers shall be black characters against a yellow background. The marking on the component B containers shall be yellow characters against a black background. The container shall be marked so that water immersion, contact with contents of container, or normal handling will not impair the legibility of the marking or remove the marking from the container. No paper lables shall be used. The marking on the containers shall include the name of the material, the type, the letter designation, the name of the manufacturer, the control (batch) number. the date of packaging, and the following instructions:

Type VII Charges

This foam powder is a tie-pan charge labeled A and B for use with type VII, dual-hopper, continuous-foam generators manufactured in accordance with Military Specification MIL-G-636.

Instructions for use:

1. Open water valve on generator.
2. Obtain proper flowing pressure as indicated on generator.
3. Open component A (yellow container) by pulling the ring attached to the strip-type cover. Pour contents into hopper either marked A or painted yellow,
4. Open component B (black container) by pulling the ring attached to the strip-type cover. Pour contents into hopper either marked B or painted black.
5. When A and B hoppers are full. open chemical control valves.

Caution:

1. Make *sure* powder is free-flowing (not caked or lumped) before emptying into hopper.
2. Powder should not be packed down into hopper. It may be pushed easily from the sides into center.
3. If powder has been allowed to remain in hopper during a temporary cessation of foam delivery, and has become wet. it may be readily flushed into the throat of the generator by means of flushout hose upon resumption of service.
4. Store charges in a cool, dry place.

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3.7 Manufacturer's records.-The control (batch) number on each container or bag shall be verifiable against the manufacturer's records of all tests made on the charges. These records shall be made available for inspection by the Government.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *For visual* examination.-Sampling for visual examination shall be in accordance with Military Standard MIL-STD-105.

4.1.2 *For acceptance testing.—Two* containers including their contents shall be selected at random from each lot.

4.2 Examination.—The preproduction sample, when required, and each sample selected at the place of manufacture in accordance with 4.1.1 shall be examined for the following defects: broken, damaged or improper container; improper fill of container; improper closure or leakage of container; improper or illegible marking; and any other requirements not requiring tests. Samplies containing one or more defects shall be considered defective. Rejection of lots shall be based on an AQL of 2.5 percent defective.

4.3 Reproduction sample.-Unless otherwise specified in the contract or order, tests of the preproduction sample will be made at the place of manufacture (see 6.2).

4.3.1 *Preparation of foam charges.*— Unless otherwise specified herein, the foam charges shall be prepared as specified in 3.6 through 3.6.3 for the type and size of charges to be used.

4.3.2 *Test conditions*.—Tests shall be performed in an atmosphere having an ambient temperature of plus 72.5° plus or minus 4.5°F., and the test apparatus shall be at the same temperature. The tests shall be performed in an area free from drafts.

4.3.3 *Fire test (for type I, sizes 1 and 2).*

4.3.3.1 *Extinguisher apparatus and charges*.—The extinguisher apparatus shall be constructed and assembled in accordance with figure 2. The body shall be capable of withstanding an internal pressure of 700 p.s.i.g. and shall be fabricated from sheet or strip steel conforming to Federal Specification QQ-S-766, class 6. The volume of the extinguisher body shall be 0.405 cubic feet plus or minus 0.002 cubic feet. The pressure relief mechanism shall be designed to open at 400 p.s.i.g. The inner container and stopple shall be in accordance with Federal Specification O-F-361. A pipe outlet 2 inches by 1/4 inch shall be attached to the top of the extinguisher. The bar stock valve shall be fabricated from stainless steel and shall have a 1/2-inch diameter valve seat opening unobstructed in the open position. The nozzle shall have a discharge channel and a straightaway channel of 3/16 inch each and finished by 1/16-inch radius to the front face and back wall of the discharge channel. The extinguisher shall be fitted with a pressure gage which is calibrated between zero and 500 p.s.i.g. The extinguisher apparatus and assembly shall be capable of withstanding an internal pressure of 500 p.s.i.g. and shall be leakproof at that pressure. For testing type I, size 1 charges, the charges shall be treated and the extinguisher charged as specified in 2.6.1.1. For testing type I, size 2 charges, the charges shall be reduced proportionally to equal charges required for type I, size 1 charges and the extinguisher shall be treated and charged as specified in 3.6.1.1.

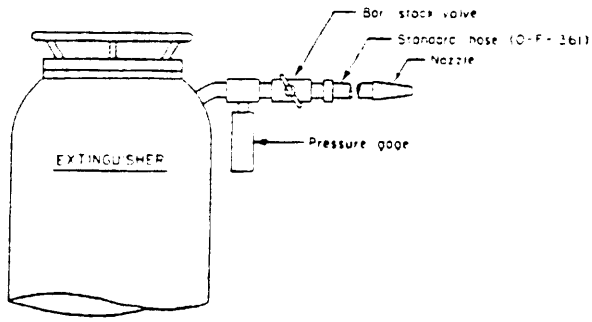


FIGURE 2.—Standard test device.

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4.3.3.2 *Procedure*.—A 4-foot diameter, cylindrical sheet steel vessel with side-walls 2 feet high shall be filled with water to a depth of 10 inches and shall be filled with 83 octane motor fuel an additional 2 inches. The gasoline shall be ignited, and 20 seconds shall elapse before the extinguisher with the charges is used as specified in 4.3.3.1. The foam stream shall be directed against the wall of the vessel and not into the burning gasoline. Attack may be made with movement of the operator permitted and initial application of foam shall be made at a distance of not less than 10 feet from the vessel. The following data shall be recorded: Type and size of charges; manufacturer; batch number; test and specification number; amount of fuel used; ambient temperature and humidity at preburn time ; time required for extinguishment, or total discharge time if no extinguishment was effected; and depth of foam blanket formed. Compare the results of the test with 3.4 and 3.5.3.3.

4.3.4 *Range, pressure, and discharge rate test (type I, sizes 1 and 2)*.—The extinguisher apparatus and charges shall be used as specified in 4.3.3.1 The maximum pressure with the nozzle shut off, the pressure after 5 seconds with the nozzle closed, the maximum pressure with the nozzle open, and the discharge rate and the range with the nozzle axis elevated 45 degrees up from the horizontal shall be determined.

4.3.5 *Foam volume test.*

4.2.5.1 Appararus and charges.

4.3.5.1.1 *Type I, size 1.*— The extinguisher apparatus and charges for type I, size 1 shall be used as specified in 4.2.3.1. The extinguisher shall be completely discharged into a container of sufficient size to hold and calibrate the total discharge volume of the extinguisher.

4.3.5.1.2 *Type I, size 2.*— The extinguisher conforming to Military Specification MIL-F-482 shall be used to test type I, size 2 foam charges, and the extinguisher shall be charged as specified in 3.6.1.2.

4.3.5.1.3 Type III.—The foam generator conforming to Military Specification MIL-G-636, type I shall be used and shall be charged as specified in 3.6.3 with natural or synthetic sea water as the solvent. The generator shall be completely discharged into a cylindrical tank or other container of sufficient size to hold and calibrate the total discharge volume of the extinguisher. The generator shall be supplied with sea water at a pressure of 50 p.s.i.g. The foam shall be discharged through 100 feet of 2½-inch, cotton-jacketed, rubber-lined fire hose conforming to Federal Specification ZZ–H-451 and a nozzle having a 1 3/4-inch tip. Synthetic sea water of the following composition shall be used if natural sea water is unavailable:

	Percent by weight
Magnesium chloride ($MgCl_2 \cdot 6H_2O$)	1.10
Calcium chloride ($CaCl_2 \cdot H_2O$)	0.16
Sodium Sulfate Anhydrous (Na_2SO_4)	0.40
Sodium chloride, C.P. (NaCl)	2.50
Fresh water (H_2O)	95.84

4.3.5.1.4 *Type VII*.—The foam generator conforming to Military Specification MIL-G-636, type II, size 1 or 2, shall be used to test type VII foam charges, and the foam generator shall be charged as specified in 3.6.3. The foam generator shall be completely discharged into a container of sufficient size to hold and calibrate the total discharge volume of the foam generator.

4.3.5.2 *Procedure.* — The foam generator shall be completely discharged into the container with the tip of the foam nozzle touching the inside wall of the container. Collecting devices such as headers, ducts, or chutes shall not be used. The volume of foam shall be measured.

4.3.5.3 *Results.*—The foam volume shall be recorded in gallons per pound of dry chemical

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foam charge used. Compare the results with 3.5.1. For type I. sizes 1 and 2, and type VII charges, volume of foam shall be determined 15 minutes after the original determination of the foam volume, and the results shall be compared with 3.5.1.1.

4.3.6 Heat resistance test.

4.3.6.1 Foam samples.—For type I, size 1, foam samples shall be taken after the test performed in 4.3.5 using foam generated in that test. For type I, size 2 and types III and VII, foam samples shall be taken after the test performed in 4.3.5 using foam generated during the middle discharge period of the extinguisher in that test.

4.3.6.2 procedure, — One-half gallon of gasoline shall be floated on water in a steel pan measuring 12 inches square by not less than 6 inches deep. A freeboard of 2 inches shall be left. The pan shall be completely filled using foam samples specified in 4.2.6.1. The foam blanket shall be levelled off to the top edge of the pan, and then an opening 3 inches wide by 12 inches long shall be made at one end of the foam blanket. The exposed fuel shall be ignited 5 minutes after the foam was first applied.

4.3.6.3 Results.—The foam blanket shall be examined after the fuel has been consumed. Compare the results of the test with 3.4 and 3.5.2.

4.3.7 *Polyethylene film tests.* — Tensile strength and elongation shall be determined by method B of Federal Specification LP-406, method 1013. Seam strength shall be determined by method A of Federal Specification L-P-406, method 1013. except that the specimen shall be taken from the seam of the bag. The seam shall be in the center of the specimen and perpendicular to the direction of pull. Water vapor permeability shall be determined by method B of Federal Specification L-P-406, method 7032. Water absorption shall be determined as specified in Federal Specification L-P-406, method 7031. Tear strength shall be determined in accordance with Federal Specification UU-P-31. method 170.

4.4 Acceptance tests. --Samples selected in accordance with 4.1.2 shall be tested as specified in 4.3.3, 4.3.4 .4.3.5 and 4.3.6. Acceptance criteria shall be the same as that specified for the preproduction sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 *Type I, size 1 charge.*— The A and B components shall be separately packaged in commercial airtight and moistureproof metal containers, or the A and B components shall be separately packaged in polyethylene bags. Each bag shall be large enough to hold the quantity required without the bag being entirely filled after sealing so that the bag will be flexible. Care shall be used in filling the bags to prevent that portion of the bag to be heat-sealed from contact with the chemical. The packaged components A and B comprising each complete charge shall be packaged together in a paperboard folding box conforming to Federal Specification PPP-B-566. The boxes shall be closed and sealed in accordance with the appendix to the box specification. Marking for identification on the boxes shall be as specified in 3.6. The bags shall conform to the requirements specified in 5.1.1.1.1.

5.1.1.1.1 *Bags*.—The bags shall be fabricated of 0.0040 inch plus or minus 0.0002 -inch-thick polyethylene film in the form of extruded lay-flat tubing without side seams, The width of the end seams of the bags shall be not less than the total thickness of 2 layers of the material, heat sealed with a smooth finish. No type of silicon agent shall be used in the fabrication of the seams. The polyethylene film shall be unfilled, virgin polyethylene of uniform average molecular weight with

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not more than 10 percent (butyl rubber and antioxidant). The film shall be uniform in color, texture, finish, and other physical properties and shall be free from porosity, warpage, checks, cracks. chipped edges, blister, scratches, and defects affecting serviceability or appearance. The film shall be transparent waxy-white in color and have a smooth surface similar to a polished finish. The film shall conform to the properties listed in table III.

TABLE III,—*Physical properties*

Property	Requirements	Test paragraph
Tensile strength (all directions) minimum p.s.i.	1500	4.3.7
Ultimate elongation nominal thickness 0.005 inch or less, minimum percent	400	4.3.7
Water absorption, maximum percent by weight	0.03	4.3.7
Toluene extractable material (77° F. plus or minus 2° F.) percent	6.0	4.3.7
Tear strength (with the direction of tear perpendicular to seam) minimum grams/mil of thickness	110	4.3.7
Seam strength, area based on thickness of single film minimum p.s.i.	1200	4.3.7
Water vapor permeability single thickness of film, 0.004 inch maximum, grams/100 sq. in/24 hours	0.45	4.3.7
Single thickness of film including 60 mm. seam, 0.004 inch maximum grams/100 sq. in/24 hours	0.52	4.3.7

5.1.1.2 *Type I, size 2* charges.—The A and B components in the quantity required for each charge shall be packaged separately in commercial airtight and moistureproof metal containers. Marking for identification shall be as specified in 3.6.

5.1.1.3 *Type III charges.* — Each charge shall be packaged in a commercial, hermetically sealed open-top steel pail. The lid shall be of the rip-off type with the sealing ring provided with a pull-off attachment. The pail shall have a bail securely attached. and the bail shall be provided with a handle. Marking for identification shall be as specifies in 3.6.

5.1.1.4 *Type VII* charges .—The A and B components shall be packaged separately in the quantity required for each charge in commercial airtight and moistureproof metal containers. Marking for identification shall be as specified in 3.6.

5.1.1.5 *Intermediate packaging.*

5.1.1.5.1 Type I, size 1 charges. Type I, size 1 charges shall be intermediate packaged in quantities of six complete charges in a fiber box conforming to Federal Specification PPP-B-636, type I, class 1, style RSC. The boxes shall be close-fitting, closed and sealed in accordance with the appendix to the box specification.

5.1.1 .5.2 *Type I, size 2* charges.—The components of each charge shall be intermediate packaged in a fiber box conforming to Federal Specification PPP-B-636, type I. class 1, style RSC. The boxes shall be close-fitting, closed and sealed in accordance with the appendix to the box specification.

5.1.1 .5.3 *Type III and type VII charges.*—Type III charges and type VII shall not require intermediate packaging.

5.1.2 Level C.—All types of charges shall be packaged as specifier in 5.1.1.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 Type I, size 1 and size 2 charges.—Complete charges of like description, intermediate packaged as specified in 5.1.1.5, shall be packed in nailed wood boxes conforming to

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Federal Specification PPP-B-621, class 2, style 2, 2½, or 3; wood cleated-plywood boxes conforming to Federal Specification PPP-B-601, overseas type, style A or B, plywood type HI, class 1; in wirebound wood boxes conforming to Federal Specification PPP-B-585, style optional, class 3; or in wood-cleated fiberboard boxes conforming to Federal Specification PPP-B-591, overseas type, style A or B. The boxes shall be close-fitting and closed and strapped in accordance with the appendix to the applicable box specification except straps shall be flat steel type I, class B. The gross weight of each box shall not exceed 200 pounds.

5.2.1.2 *Type III* charges.-Type III charges packaged as specified in 5.1.1.3 shall not require overpacking.

5.2.1.3 *Type VII charges.* — Complete charges packaged as specified in 5.1.1.4 shall be packed in a container in the manner specified in 5.2.1.1.

5.2.2 Level B.

5.2.2.1 Type I, size 1 and size 2 charges.— The charges of like description intermediate packaged as specified in 5.1.1.5 shall not require overpacking.

5.2.2.2 *Type III* charges.—Type III charges packaged as specified in 5.1.1.3 shall not require overpacking.

5.2.2.3 *Type VII charges.* — Complete charges packaged as specified in 5.1.1.4 shall be packed in nailed wood boxes conforming to Federal Specification PPP-B-621, class 1, style 2, 2½, or 3; wood cleated-plywood boxes conforming to Federal Specification PPP-B-601, domestic type, style as applicable, plywood type IV, class 1; wirebound wood boxes conforming to Federal Specification PPP-B-586, style optional class 1 or 2; or wood-cleated fiberboard boxes conforming to Federal Specification PPP-B-591, domestic-type style optional. The boxes shall be closefitting and closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of each box shall not exceed 200 pounds.

5.2.3 *Level C.*

5.2.3.1 *Type I, size 1 and size 2 charges.*— Complete charges of like description intermediate packaged as specified in 5.1.1.5.1 and 5.1.1.5.2 shall not require overpacking.

5.2.3.2 *Type III charges.* — Complete charges packaged as specified in 5.1.1.3 shall not require overpacking.

5.2.3.3 Type VII charges. — Complete charges packaged as specified in 5.1.1.4 shall be packed in containers complying with Consolidated Freight Classification Rules and Container Specifications or with National Motor Freight Classification Rules and Container Specifications or with other carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military.—In addition to any special marking required, packages and shipping containers shall be packed in accordance with Military Standard MIL-STD-129.

5.3.2 Nonmilitary.

5.3.2.1 Interior containers.—Packages and interior containers shall be marked with the name of the material and the name or trademark of the contractor, of such known character that the source of supply may be readily determined.

5.3.2.2 *Exterior* containers.-Shipping containers shall be marked with the name of the item, the type and class, as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

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6. NOTES

(Nonmilitary agencies. —Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

6.1 Intended use.—Type I charges are intended for 2½-gallon portable extinguisher (size 1 only) conforming to Federal Specification O-F-361 and the nominal 40-gallon wheeled extinguishers (size 2 only) conforming to Military Specification MIL-F-482. Type III charges are intended for continuous chemical foam single-hopper generators conforming to Military Specification MIL-G-636, type I. Type VII charges are intended for continuous chemical foam, dual hopper generators conforming to Military Specification MIL-G-636, type II, sizes 1 and 2.

6.2 Ordering data. — Procurement documents should specify the following:

((2) Title, number, and date of this specification.

(b) Type and size required (see 1.2.1).

(c) If preproduction sample is required, and where tests will be conducted (see 3.2 and 4.3).

(d) Level of packaging and packing required (see section 5).

(e) Marking, if different (see 5.3).

6.3 Supersession data.—The following is a comparison between the types and sizes of this specification and the types of cancelled Military Specification JAN-C-267:

This specification	*JAN-C-267*
type I, size 1	type 1
type I, size 2	type 6
type III	type 3
type VII	type 7

6.4 Transportation description.-Transportation description applicable to this item is:

Fire extinguisher charges or compounds, not otherwise indexed by name.

Carload minimum. weight 36,000 pounds.

Motor volume minimum weight 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army-E T
Navy-Y A MC S.

Copies of this specification may be purchased for 10 cents each.

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